Emotional Speech Synthesis via Visual Context Perception

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Abstract. In the current era of artificial intelligence (AI) innovation, the perceptual mapping between vision and speech plays a crucial role in facilitating multi-modal interactions. While extensive research has discovered the semantic relationship between speech and image, studies on modeling emotional mapping are relatively scarce, even though semantics and emotion can be equally important. Our research explores emotional speech synthesis via visual context perception. We first propose a novel vision-based emotional text-to-speech (VE-TTS) model for a new task that aims to utilize implicit emotional information to directly control the emotional expression of synthesized speech. Experiments demonstrate that our model can not only effectively synthesize emotional speech to enhance the user experience, but also has great potential for applications in various other technologies. Additionally, we introduce a multi-modal contextual speech dataset for benchmarking this task. Finally, a user study on blind or visually impaired (BLV) users also demonstrates the potential of emotional speech in various areas.

Keywords: Multi-modality · Emotional Speech Synthesize · AIGC.

1 Introduction

In the current era of AI innovation, the perceptual mapping between vision and speech plays a crucial role in enhancing user experience, improving emotional expression, and facilitating multi-modal interactions. The advent of the GPT-4o[3] allows visual and vocal modalities to seamlessly integrate, catalyzing substantial advancements and garnering considerable scholarly attention [6]. As shown in Fig. 2, its applications can span various domains, enhancing user experiences by combining different modalities, such as vision and speech, to create more immersive and intuitive interactions. The multi-modal mapping process is vital for providing feedback based on recognizing and understanding real-life context, because our cognition of the world is often multi-modal [10].

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[3] https://openai.com/index/hello-gpt-4o/

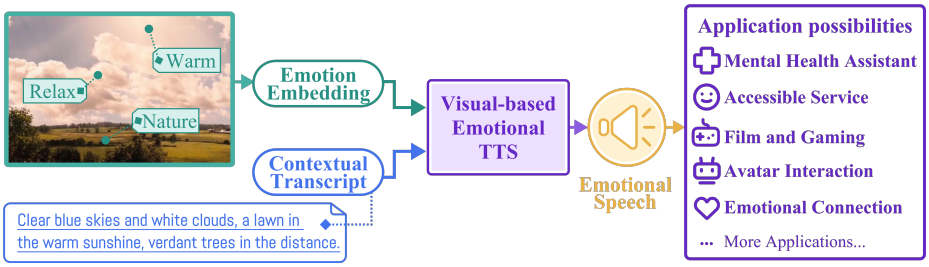


Fig. 1. Overview of the emotional speech synthesize model's workflow and future possibilities. The multi-modal interaction may not only well support the technical applications but also may assist more extensive application scenarios.

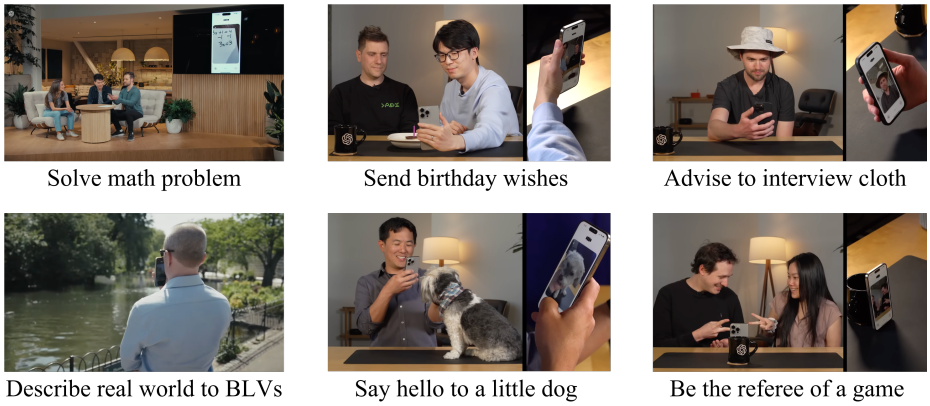


Fig. 2. Interaction of GPT-4o with real-life camera inputs[4]

However, despite the extensive research on semantic mapping between speech and image, e.g., video content captioning and video content retrieval [30,33,29], we argue that the inclusion of emotional context is equally indispensable. The emotional information embedded in visual context can be effectively conveyed through emotionally expressive speech, thereby enhancing the vividness and relevance of the speech [24]. However, existing image-controlled speech synthesis models tend to focus on human facial features in an image, such as gender and age [14,13,21]. While these features are indeed important, such a narrow focus tends to overlook the rich contextual details present in the overall image, including color, lighting, spatial context, symbolic objects, and stylistic cues. These contextual details are crucial for a comprehensive and accurate representation of emotion, as they can convey the full emotional tone of a scene [23].

Therefore, our research focuses on this critical aspect, emphasizing the intricate emotional mapping between speech and image. The analysis of contextual details in images enables the control and synthesis of emotionally expressive speech in an implicit and context-driven manner. Studies have demonstrated that non-facial features, such as the background environment, can effectively

[4] https://openai.com/index/hello-gpt-4o/

complement facial expressions, providing a more comprehensive emotional expression [23]. One of the principal challenges confronting our work is the scarcity of suitable datasets containing multi-modal information. To address this issue, we have developed a multi-modal contextual-speech dataset in collaboration with generative AI, encompassing the three modalities of speech, text, and vision.

Our contributions are threefold:

1. **New task in vision-vocal interaction.** We introduce a task, that is to support applications similar to GPT-4o, utilizing emotional mapping between speech and image may be crucial.
2. **Vision-based emotional TTS model.** We propose the VE-TTS model, designed to effectively map emotions between speech and image and to synthesize emotionally expressive speech.
3. **Multi-modal contextual speech dataset.** We construct a multi-modal dataset containing speech, text, and vision, serving as a foundational resource for studying the relationship between speech and image in emotional contexts.

2 Related Works

Speech primarily delivers semantics, while the emotion revealed in speech can enhance its persuasiveness and engagement [3,4], creating a unique bond that resonates with the audience [26]. Although methods exist to extract semantic information from visual content, such as video and image [7], studies on exploiting emotional mappings between visual and vocal information seem to be relatively lacking [10,28]. A study in color psychology proposed a color-image coordinate system based on people's emotional responses to colors. Colors are divided into four general categories soft, hard, cool, and warm. Each category can invoke vastly different emotions. For example, bright red is associated with gentleness whereas dark black-brown matches with the feeling of thick and heavy [19]. It suggests that images may contain valuable emotional information.

The current emotional TTS models mainly control emotion embedded in a synthesized speech via a discrete "setting", which may be a limited number of emotional tags or natural language descriptions [2]. For instance, PromptTTS provides methods to integrate natural language instructions [14,22], and InstructTTS can interpret natural language descriptions to synthesize stylistic speech [35]. This is contrary to how acoustic information is perceived by human multi-sensory systems in reality, thereby reducing its effectiveness in imitating emotion during speech synthesis [10].

Some vision-based TTS can model the relationship between human-centric input images and the synthesized speech [17]. For example, Yang et al. focused on generating speech from 3D face shapes [36], while MM-TTS can utilize emotional cues from facial images to control the style of the generated speech [13]. Semantic mapping between vision and vocal enables the integration of complementary human senses, improving cognitive performance and resulting in more expressive and emotionally accurate synthesized speech [32]. For instance, Evans

suggests that certain visual features naturally align with auditory features, and thus may improve task performance [11]. Moreover, previous works have proposed that emotion expression performance can be enhanced by integrating additional modals [1,34]. This emphasizes the necessity for multi-modal integration to incorporate different sensory inputs such as speech and vision.

However, research in vision-based TTS concentrates on human portraits, including facial expressions, body forms, and gestures [12,13,21,27,36].Little attention has been paid to the background environment, which may contain vital contextual information. Incorporating environmental context allows the model to better interpret and understand nuanced emotions in speech, thereby substantially improving the realism and emotional accuracy of synthesized speech [23,16]. One application is the transferable speech-visual text generation (TAVT) framework, which uses a speech-visual meta-mapping network to significantly enhance the generation of contextually rich and emotionally engaging text from video inputs [29,25].

3 Multi-Modal Contextual Speech Dataset

Multi-modal sentiment datasets are lacking from the current study, to construct a multi-modal contextual speech dataset, it is essential to obtain speech data that is emotionally expressive or strongly correlates with the image. For this reason, we collect audiobook data from the internet. Due to its uni-modal characteristics, descriptions of the scene context in audiobook speech are utilized in this study.

3.1 Data Construction

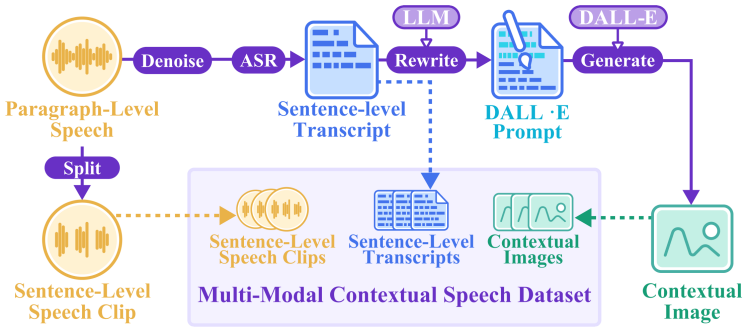


Fig. 3. Multi-modal contextual speech dataset

The construction pipeline for our dataset is illustrated in Fig. 3.1. We used an automatic speech recognition (ASR) system [31] to convert the enhanced

discourse-level speech into sentence-level transcripts. Then we employed GPT-3.5 turbo [5] and GPT-4o to optimize the transcripts, addressing issues of inconsistency and varying data quality[5].

The sentence-level transcripts are then summarized and transformed into DALL-E prompts with the assistance of an LLM for the next step.

Finally, the DALL-E model is employed for text-to-image generation. After multiple rounds of image generation attempts, the generated contextual images accurately represent the described scenes and convey the intended emotional tone. The visual accuracy, emotional relevance, and consistency with the corresponding textual descriptions remain relatively stable across the batch-generated images.

3.2 Data Analysis

Fig. 4. Dataset samples of images and transcripts.

We select horror stories, children's stories, and game commentaries because these types of speech exhibit strong emotional characteristics. We collect over 200 hours of raw speech from the internet, and the final effective sentence-level speech clips used for speech reconstruction training encompass over 103 hours. These sentence-level speech clips constitute the modality data of the speech in the dataset. The text modality data consists of approximately 8,261 sentence-level transcripts. Using DALL-E, we obtain 3376 images, which comprise the vision modality data. Some image and transcript pair samples from our dataset are shown in Fig. 4.

[5]These sentence fragments are analyzed to identify, select, and integrate sentences, forming sentence-level transcripts that serve as a reference for splitting paragraph-level speech into sentence-level speech clips of the appropriate length.

4 Emotional Speech Synthesize via Visual Context Perception

As illustrated in Fig. 5, our VE-TTS framework consists of a CLIP image encoder, a projection layer as an adaptor, and a VITS-based speech backbone. The VITS-based speech backbone is an end-to-end variational autoencoder including a reference speech encoder and a speech decoder. The speech encoder comprises a stack of convolutional neural network layers and a gated recurrent unit (GRU) layer, which are responsible for learning global style embeddings from mel spectrograms of ground truth speech. The image encoder extracts global visual embeddings from the input image.

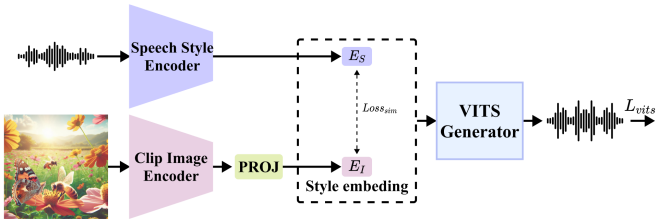


Fig. 5. Overview of VE-TTS training framework.

4.1 Model Training

Given the necessity of end-to-end speech-image alignment, which relies on a large amount of strongly correlated speech-image data, a pre-training fine-tuning training method is employed.

To enhance the robust speech modeling capabilities, the VITS-based speech backbone is initially trained in an unsupervised manner on a large corpus of emotional speech data. The speech embedding is utilized as the style embedding for the VITS generator. We employ multiple discriminators for adversarial training. Following normalization, the global style embedding is transmitted to the generator, which oversees the training of several modules, including the post encoder, prior encoder, and duration predictor [18,20]. For simplicity, we label the training loss of the backbone as c, which consists of multiple adversarial loss and reconstruction losses.

During the fine-tuning stage, we employ the frozen reference speech encoder as supervision for the unfrozen image encoder. The image embedding is used as the style embedding for the VITS generator. We directly use cosine similarity loss $\mathcal{L}_{sim}$ to achieve alignment. To further ensure the effectiveness of the model, we apply the same loss $\mathcal{L}_{vits}$ to guide the alignment process.

4.2 Implementation Details

For the speech backbone, we adopt the design of Bert-VITS2[5]. To improve rhythmicity, it uses the phonemes and text features from the pre-trained BERT model [15,8,9] as inputs to the speech decoder, and uses multi-discriminators as the stream model and duration predictor [20] inputs. We replace the stochastic duration predictor with a normal duration predictor and enable the gradient of the duration predictor for global speech embedding during backpropagation. To utilize the cosine similarity, we apply normalization on the channel dimension of style embeddings. For the CLIP image encoder, we use the open-source CLIP-ViT-H-14. During the fine-tuning stage, the reference speech and the generator are frozen, while the last six transformer layers and the projection layer of the CLIP encoder, as well as discriminators, are unfrozen.

5 Experiments

This section presents the results of experiments conducted on the synthesis of emotional speech via visual context perception. The experiments included subjective evaluation experiments on speech-image emotion consistency and a user study on BLVs. The results of these experiments have proven the existence of emotional mapping between speech and image, the ability of VE-TTS to synthesize emotional speech, and the potential for the application of synthesized emotional speech.

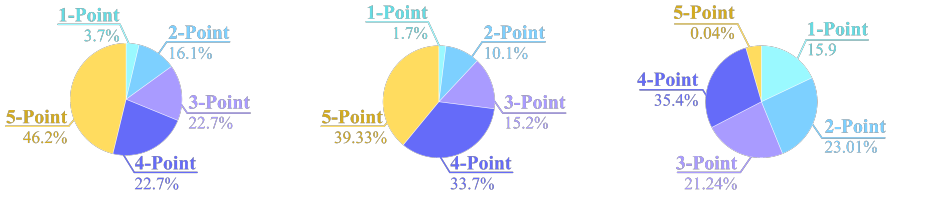
5.1 Emotion Consistency of Speech and Image

We apply subjective experiments on three sets of speech-image pairs: (1) Speech and images in videos collected from the Internet, (2) Emotional speech synthesis via visual context perception in images by VE-TTS, (3) Unemotional speeches synthesized based on the semantic synthesis of image content only, and images. Ten participants are invited to complete a subjective assessment using a five-point Likert scale. Descriptive statistics are calculated using mean calculation and relative frequency distribution.

The scores for the three data pairs are counted separately for comparative analysis (as shown in Fig. 6). The results indicate that the mean consistency score for the video speech and image pairs suggests a strong overall consistency in the first set. Meanwhile, the distribution pattern of the emotional speech and image pairs ratings closely mirrors that of the video speech and image pairs, that is, the emotional speech generated by VE-TTS performs similarly to video speeches in terms of emotional mapping between speech and image.

An emotion level mapping between speech and image can indeed be established, and our VE-TTS synthesized emotional speech can reproduce the image emotion better, while the emotionless only semantically related speech is missing this part.

[5] https://github.com/fishaudio/Bert-VITS2



(A) Video speeches and images (B) Emotional speeches and images (C) Unemotional speeches and images

Fig. 6. Relative frequency distribution of scores for the three speech-image pairs. As illustrated in the chart, the distribution of scores shows a similarity in the video speech and image pairs and the emotional speech and image pairs.

5.2 User Study on BLVs

An additional subjective experiment is conducted with five BLV participants, to test emotional speeches in the application of accessible AD. BLV participants are provided with only unemotional speeches and emotional speeches and are required to infer the contextual style of the stories. A post-test opinion collection is conducted to collect subjective opinions.

The inference results and post-test opinions show that emotional speech synthesis has a profound impact on enhancing the user experience of synthesized accessible AD, which is one of the potential application possibilities.

5.3 Experiment Conclusion

In a nutshell, several experiments prove that the association and mapping between speech and image can indeed be established at the emotional level, and the speech synthesized by VE-TTS can better reproduce the emotion of the image. Compared to unemotional synthesized speech, emotionally synthesized speech can indeed provide a better experience to the listener, and it has great potential for various applications.

6 Conclusion

This paper investigates emotional speech synthesis through visual context perception, particularly focusing on the mapping of emotions between speech and image. We confirm that a resolvable emotion mapping exists between speech and image, Experimental verification demonstrates the proposed VE-TTS framework can reasonably and efficiently infer this mapping mechanism. To support our research, we have developed a multi-modal contextual speech dataset that includes data from three modalities: speech, text, and vision. The dataset is a valuable resource for advancing research in this area, and the construction method is also applicable to other areas that require detailed and expressive speech descriptions, such as AD generation for BLVs. Overall, our research demonstrates the potential for incorporating visual emotion context into speech synthesis, thereby enhancing multi-modal interactions to support applications similar to GPT-4o.

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